Kubernetes – 02

Create the YAML file

On your master node:

```
apiVersion: v1
kind: Pod
metadata:
  name: firstpod
spec:
  containers:
    - name: nginx
      image: nginx
```

Create the Pod

kubectl create -f firstpod.yml

```
root@master:~# kubectl create -f firstpod.yml
pod/firstpod created
```

```
root@master:~# kubectl create -f firstpod.yml
pod/firstpod created
root@master:~# kubectl get pods
NAME       READY   STATUS    RESTARTS      AGE
firstpod   1/1     Running   0             27s
nginx      1/1     Running   1 (30m ago)   41h
root@master:~#
```

```
root@master:~# kubectl logs firstpod
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/11/12 06:50:05 [notice] 1#1: using the "epoll" event method
2026/11/12 06:50:05 [notice] 1#1: nginx/1.31.8
2026/11/12 06:50:05 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/11/12 06:50:05 [notice] 1#1: OS: Linux 7.0.0-1014-aws
2026/11/12 06:50:05 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/11/12 06:50:05 [notice] 1#1: start worker processes
2026/11/12 06:50:05 [notice] 1#1: start worker process 29
2026/11/12 06:50:05 [notice] 1#1: start worker process 30
root@master:~#
```

Task 2 — Set Environment Variables in a Pod.

```
apiVersion: v1
kind: Pod
metadata:
  name: firstpod-env
spec:
  containers:
    - name: nginx
      image: nginx
      env:
        - name: myname
          value: "sabair"
        - name: City
          value: "Hyderabad"
```

Check the environment variables

kubectl exec firstpod-env – env

```
root@master:~# kubectl get pods
NAME           READY   STATUS    RESTARTS      AGE
firstpod       1/1     Running   0             2m46s
firstpod-env   1/1     Running   0             10s
nginx          1/1     Running   1 (32m ago)   41h
root@master:~# kubectl exec firstpod-env -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=firstpod-env
NGINX_VERSION=1.31.8
NJS_VERSION=1.0.2
NJS_RELEASE=1~trixie
ACME_VERSION=0.4.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
myname=sabair
City=Hyderabad
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
KUBERNETES_SERVICE_HOST=10.96.0.1
HOME=/root
root@master:~#
```

Task 3 — Pod with Commands (args) in YAML

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: firstpod-args
spec:
  containers:
    - name: nginx
      image: nginx
      command: ["/bin/sh"]
      args: ["-c", "sleep 50"]
```

Create the pod

```
root@master:~# kubectl get pods
NAME           READY   STATUS    RESTARTS      AGE
firstpod       1/1     Running   0             5m20s
firstpod-args  1/1     Running   0             9s
firstpod-env   1/1     Running   0             2m44s
nginx          1/1     Running   1 (35m ago)   42h
root@master:~#
```

Verify the command and args - kubectl describe pod firstpod-args

```
Name:             firstpod-args
Namespace:        default
Priority:         0
Service Account:  default
Node:             worker-01/10.30.1.102
Start Time:       Thu, 12 Nov 2026 06:55:11 +0000
Labels:           <none>
Annotations:      cni.projectcalico.org/containerID: b4f6e590bafa1052b23a160ac833fd156c2dfcf368d162f65ac68de1b2cdddf8
                  cni.projectcalico.org/podIP: 192.168.171.2/32
                  cni.projectcalico.org/podIPs: 192.168.171.2/32
Status:           Running
IP:               192.168.171.2
IPs:
  IP:  192.168.171.2
Containers:
  nginx:
    Container ID:  containerd://1fe0d1bdc9d674a68c07e844c4e9a40a148a88960ce5992611d881237abad7e1
    Image:         nginx
    Image ID:      docker.io/library/nginx@sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57e0fe56caf67512486aff76d82
    Port:          <none>
    Host Port:     <none>
    Command:
      /bin/sh
    Args:
      -c
      sleep 50
    State:          Running
      Started:      Thu, 12 Nov 2026 06:55:12 +0000
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-xw6sb (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-xw6sb:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
```

Task 4 — Create a Pod with Two Nginx Containers

Create the YAML

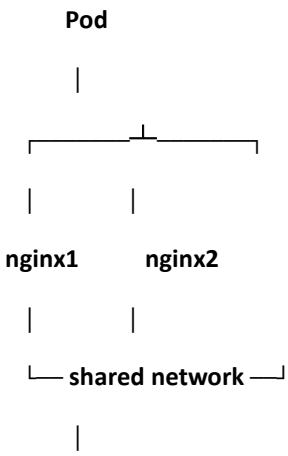
```yaml
apiVersion: v1
kind: Pod
metadata:
  name: two-nginx-pod
spec:
  containers:
    - name: nginx1
      image: nginx

    - name: nginx2
      image: nginx
      command: ["/bin/sh"]
      args:
        - -c
        - |
          sed -i 's/listen       80;/listen       8080;/' /etc/nginx/conf.d/default.conf
          nginx -g 'daemon off;'
```

Create the Pod - kubectl create -f two-nginx-pod.yml

```
root@master:~# kubectl create -f two-nginx-pod.yml
pod/two-nginx-pod created
root@master:~# kubectl get pods
NAME            READY   STATUS    RESTARTS       AGE
firstpod        1/1     Running   0              10m
firstpod-args   1/1     Running   4 (105s ago)   5m46s
firstpod-env    1/1     Running   0              8m21s
nginx           1/1     Running   1 (40m ago)    42h
two-nginx-pod   2/2     Running   0              7s
root@master:~# kubectl exec two-nginx-pod -c nginx1 -- curl -I http://localhost:8080
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0   896    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
HTTP/1.1 200 OK
Server: nginx/1.31.8
Date: Thu, 12 Nov 2026 07:01:15 GMT
Content-Type: text/html
Content-Length: 896
Last-Modified: Wed, 21 Oct 2026 14:24:51 GMT
Connection: keep-alive
ETag: "6ad8cb33-380"
Accept-Ranges: bytes
```

confirms that `nginx1` successfully communicated with `nginx2` through `localhost:8080`.



Both containers share the same Pod IP/network namespace.

Task 5 — Init Container

- **An init container that sleeps for 30 seconds.**
- **A main Nginx container that starts only after the init container finishes.**

Create YAML

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: initpod
spec:
  initContainers:
    - name: init-container
      image: busybox
      command: ["sh", "-c", "echo Init container started; sleep 30; echo Init container completed"]

  containers:
    - name: nginx
      image: nginx
```

Create the Pod

```
root@master:~# kubectl create -f initpod.yml
pod/initpod created
root@master:~# kubectl get pods
NAME            READY   STATUS             RESTARTS      AGE
firstpod        1/1     Running            0             13m
firstpod-args   0/1     CrashLoopBackOff   5 (36s ago)   8m43s
firstpod-env    1/1     Running            0             11m
initpod         0/1     Init:0/1           0             7s
nginx           1/1     Running            1 (43m ago)   42h
two-nginx-pod   2/2     Running            0             3m4s
root@master:~#
```

```
NAME            READY   STATUS             RESTARTS      AGE
firstpod        1/1     Running            0             14m
firstpod-args   0/1     CrashLoopBackOff   5 (64s ago)   9m11s
firstpod-env    1/1     Running            0             11m
initpod         1/1     Running            0             35s
nginx           1/1     Running            1 (44m ago)   42h
two-nginx-pod   2/2     Running            0             3m32s
root@master:~#
```

initpod 1/1 Running

```
root@master:~# kubectl logs initpod -c init-container
Init container started
Init container completed
root@master:~#
```

Task 6 — Generate YAML using `kubectl run --dry-run`

Generate the YAML- kubectl run nginx-dryrun --image=nginx --dry-run=client -o yaml

```
root@master:~# kubectl run nginx-dryrun --image=nginx --dry-run=client -o yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: nginx-dryrun
  name: nginx-dryrun
spec:
  containers:
  - image: nginx
    name: nginx-dryrun
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
root@master:~#
```

Instead of copying it manually, use:

kubectl run nginx-dryrun --image=nginx --dry-run=client -o yaml > nginx-dryrun.yml

Check the file:

```
root@master:~# kubectl run nginx-dryrun --image=nginx --dry-run=client -o yaml > nginx-dryrun.yml
root@master:~# cat nginx-dryrun.yml
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: nginx-dryrun
  name: nginx-dryrun
spec:
  containers:
  - image: nginx
    name: nginx-dryrun
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
root@master:~#
```

Modify the generated YAML

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: nginx
  name: nginx-dryrun
spec:
  containers:
  - image: nginx
    name: nginx-dryrun
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
```

Under `metadata`, add a label:

metadata:

labels:

app: nginx

name: nginx-dryrun

```
root@master:~# kubectl create -f nginx-dryrun.yml
pod/nginx-dryrun created
root@master:~# kubectl get pods
NAME           READY   STATUS    RESTARTS        AGE
firstpod       1/1     Running   0               22m
firstpod-args  1/1     Running   7 (5m16s ago)   16m
firstpod-env   1/1     Running   0               19m
initpod        1/1     Running   0               8m22s
nginx          1/1     Running   1 (51m ago)     42h
nginx-dryrun   1/1     Running   0               7s
two-nginx-pod  2/2     Running   0               11m
root@master:~# kubectl get pod nginx-dryrun --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
nginx-dryrun  1/1     Running   0          17s   app=nginx
root@master:~#
```

Task 7 — `kubectl create` vs `kubectl apply`

Create the YAML

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: create-apply-pod
spec:
  containers:
    - name: nginx
      image: nginx:1.27
```

Deploy using `kubectl create`

Check the current image

```
root@master:~# kubectl get pods
NAME           READY   STATUS    RESTARTS        AGE
firstpod       1/1     Running   0               22m
firstpod-args  1/1     Running   7 (5m16s ago)   16m
firstpod-env   1/1     Running   0               19m
initpod        1/1     Running   0               8m22s
nginx          1/1     Running   1 (51m ago)     42h
nginx-dryrun   1/1     Running   0               7s
two-nginx-pod  2/2     Running   0               11m
root@master:~# kubectl get pod nginx-dryrun --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
nginx-dryrun  1/1     Running   0          17s   app=nginx
root@master:~# nano create-apply.yml
root@master:~# kubectl create -f create-apply.yml
pod/create-apply-pod created
root@master:~# kubectl get pod create-apply-pod
NAME               READY   STATUS    RESTARTS   AGE
create-apply-pod   1/1     Running   0          15s
root@master:~# kubectl get pod create-apply-pod -o jsonpath='{.spec.containers[0].image}'
nginx:1.27root@master:~#
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: create-apply-pod
spec:
  containers:
    - name: nginx
      image: nginx:1.28
```

change t0 1.28

Try create again

```
root@master:~# kubectl create -f create-apply.yml
Error from server (AlreadyExists): error when creating "create-apply.yml": pods "create-apply-pod" already exists
```

That's an important difference:

kubectl create is intended to create a new resource. It won't update an existing Pod.

Now try `kubectl apply`

```
root@master:~# kubectl apply -f create-apply.yml
Warning: resource pods/create-apply-pod is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is
only be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotat
pod/create-apply-pod configured
```

That means `kubectl apply` successfully applied your modified YAML.

Verify the image

kubectl get pod create-apply-pod -o jsonpath='{.spec.containers[0].image}'

```
root@master:~# kubectl get pod create-apply-pod -o jsonpath='{.spec.containers[0].image}'
nginx:1.28root@master:~#
```

Remember the difference

`kubectl create -f`

- **Creates the resource.**
- **If it already exists → `AlreadyExists` error.**
- **Mainly imperative creation.**

`kubectl apply -f`

- **Creates or updates based on the YAML configuration.**
- **Maintains the last-applied configuration.**

- **Commonly used for declarative Kubernetes management.**

Task 8 — Edit an Existing Pod

use your `create-apply-pod` and change its environment variable using `kubectl edit`.

Add the environment variable directly under `name: nginx`:

- image: nginx:1.28

imagePullPolicy: IfNotPresent

name: nginx

env:

- name: myname

value: "sabair"

Task 9 — Expose `firstpod` Using a ClusterIP Service

```
apiVersion: v1
kind: Service
metadata:
  name: firstpod-service
spec:
  selector:
    app: firstpod
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: ClusterIP
```

```
root@master:~# nano firstpod-service.yml
root@master:~# kubectl get pod firstpod --show-labels
NAME       READY   STATUS    RESTARTS   AGE   LABELS
firstpod   1/1     Running   0          42m   <none>
root@master:~#
```

Because `firstpod` already exists, the easiest approach is to delete and recreate it:

```
root@master:~# kubectl create -f firstpod.yml
pod/firstpod created
root@master:~# kubectl create -f firstpod-service.yml
service/firstpod-service created
root@master:~# kubectl get svc
NAME               TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
firstpod-service   ClusterIP   10.108.44.61   <none>        80/TCP    7s
kubernetes         ClusterIP   10.96.0.1      <none>        443/TCP   43h
root@master:~#
```

Verify that the Service has an endpoint

```
root@master:~# kubectl create -f firstpod.yml
pod/firstpod created
root@master:~# kubectl get pod firstpod --show-labels
NAME       READY   STATUS    RESTARTS   AGE   LABELS
firstpod   1/1     Running   0          4s    app=firstpod
root@master:~# kubectl get endpoints firstpod-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME               ENDPOINTS           AGE
firstpod-service   192.168.37.198:80   2m47s
root@master:~#
```

Service selector:

app=firstpod

↓

✅

Pod:

app=firstpod

↓

Nginx :80

Task 10 — Pod with Resource Requests and Limits

```
apiVersion: v1
kind: Pod
metadata:
  name: resource-pod
spec:
  containers:
    - name: nginx1
      image: nginx
      resources:
        requests:
          cpu: "100m"
          memory: "64Mi"
        limits:
          cpu: "200m"
          memory: "128Mi"

    - name: nginx2
      image: nginx
      command: ["/bin/sh"]
      args:
        - -c
        - |
          sed -i 's/listen       80;/listen       8080;/' /etc/nginx/conf.d/default.conf
          nginx -g 'daemon off;'
      resources:
        requests:
          cpu: "100m"
          memory: "64Mi"
        limits:
          cpu: "200m"
          memory: "128Mi"
```

Create the yaml file

- **100m CPU = 0.1 CPU requested**
- **200m CPU = maximum 0.2 CPU**
- **64Mi = minimum requested memory**
- **128Mi = maximum memory**

Create the Pod

```
root@master:~# nano resource-pod.yml
root@master:~# kubectl create -f resource-pod.yml
pod/resource-pod created
root@master:~# kubectl get pod resource-pod
NAME           READY   STATUS    RESTARTS   AGE
resource-pod   2/2     Running   0          8s
root@master:~#
```

Verify resources

```
Status:           Running
IP:               192.168.171.9
IPs:
  IP:  192.168.171.9
Containers:
  nginx1:
    Container ID:   containerd://d142e69f1dc9ba381cf10a32c82a46b765207086a43bf1fc367dc3dc48b44d04
    Image:          nginx
    Image ID:       docker.io/library/nginx@sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57e0fe56caf67512486aff76d82
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Thu, 12 Nov 2026 07:40:01 +0000
    Ready:          True
    Restart Count:  0
    Limits:
      cpu:     200m
      memory:  128Mi
    Requests:
      cpu:        100m
      memory:     64Mi
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-995xw (ro)
  nginx2:
    Container ID:  containerd://9611bb90402ab45c2ae6ad6adab4d3d5ad60cf1d24d24861c08882d28a85217b
    Image:         nginx
    Image ID:      docker.io/library/nginx@sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57e0fe56caf67512486aff76d82
    Port:          <none>
    Host Port:     <none>
    Command:
      /bin/sh
    Args:
      -c
      sed -i 's/listen       80;/listen       8080;/' /etc/nginx/conf.d/default.conf
      nginx -g 'daemon off;'

    State:          Running
      Started:      Thu, 12 Nov 2026 07:40:01 +0000
    Ready:          True
    Restart Count:  0
    Limits:
```